

Design and Construction of a Path Detector Using NE555 Timer, LM324N Comparator, and AND Gate Logic

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Abstract—This paper presents the design, implementation, and performance evaluation of a low-cost optical path detector intended for use as the sensing front-end of line-following robotic platforms. The system distinguishes between a black line and a white background using two Light Dependent Resistor (LDR) based optical channels. An NE555 timer operating in astable mode periodically pulses the emitter LEDs, an LM324N quad operational amplifier is configured as a dual-channel voltage comparator to evaluate reflected-light levels against adjustable reference thresholds, and an MC74F08 quad 2-input AND gate combines the two comparator outputs to drive an audible buzzer only when both sensing channels simultaneously indicate an off-path condition. This logical AND combination allows the system to distinguish between partial deviation, in which only a visual LED indicator is activated, and complete path loss, in which both LED and audible alarms are triggered. The circuit was prototyped on a breadboard, verified functionally, and subsequently transferred to a printed circuit board (PCB) for a more robust implementation. Experimental testing showed reliable discrimination between black and white surfaces on both straight and curved paths, with a nominal +5 V logic supply derived from a 9 V battery via a resistive voltage-drop network. The design demonstrates that dependable optical line-sensing can be achieved using discrete analog and digital ICs, without a microcontroller, at very low component cost, making it a suitable educational platform for introducing first-year electrical and electronics engineering students to sensor interfacing, comparator design, and combinational logic.

Index Terms—path detector, line follower, NE555 timer, light dependent resistor, LM324 comparator, AND gate, astable multivibrator, optical sensing

I. INTRODUCTION

A. Background and Motivation

Line-following robots are among the most widely used introductory platforms in robotics education, since they combine analog sensor interfacing, signal conditioning, and decision logic in a single, visually verifiable system. For students in their first year of an Electrical and Electronics Engineering programme, however, the practical realization of even simple electrical laws – Ohm’s Law, the Superposition Theorem, and Norton’s Theorem – is often unfamiliar, as is the process of identifying, testing, and interconnecting real components on a breadboard. Structured hands-on projects of this kind help

bridge the gap between theoretical coursework and applied circuit design.

The specific objective of this project is to design and construct a low-cost device that can reliably sense the difference between a black path and a white background, so that it can subsequently be used as the sensing module of a line-following robot capable of tracking both straight and curved lines. Cost, component availability, and ease of construction were treated as primary design constraints, motivating a discrete-IC approach rather than a microcontroller-based solution.

B. System Overview

The system is organized into three cooperating functional stages: (i) an NE555 timer configured in astable mode, which generates a continuous pulse train that drives the emitter LEDs; (ii) two LDR/potentiometer voltage dividers whose outputs are evaluated by an LM324N operational-amplifier comparator stage against a fixed reference voltage; and (iii) an MC74F08 AND gate that combines the two comparator outputs to drive an alarm buzzer, while each comparator output independently drives a dedicated indicator LED. The complete PCB-level realization of this system is shown in Fig. 1, and the assembled breadboard prototype used for functional verification is shown in Fig. 2.

C. Schematic Parts Reference

Table I lists the components used in the construction of the path detector.

TABLE I
SCHEMATIC PARTS REFERENCE

S.No	Name	Qty	Description
1	Breadboard	1	–
2	Resistor	6	1K, 480K, 1.2K, 1.5K
3	Capacitor	1	1000 μ F
4	Potentiometer	2	100 K
5	LED	3	–
6	Timer	1	NE555
7	Comparator	1	LM324N
8	AND Gate	1	MC74F08
9	LDR	1	–

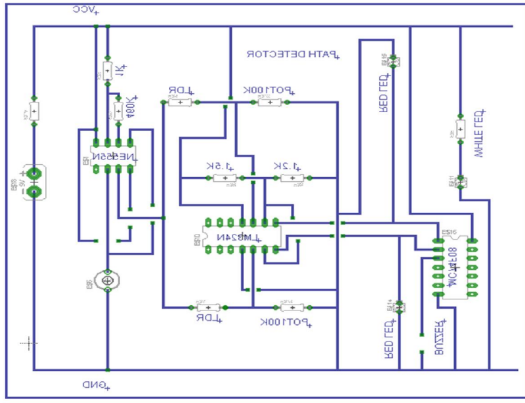


Fig. 1. PCB board overview of the path detector, showing the NE555 timer, LDR/potentiometer sensing network, LM324N comparator, MC74F08 AND gate, indicator LEDs, and buzzer.

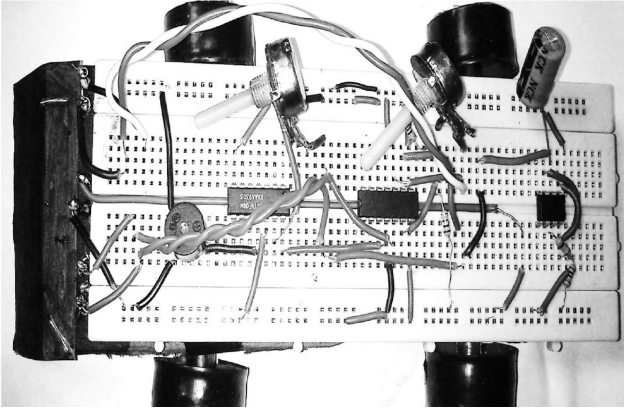


Fig. 2. Breadboard circuit arrangement used for functional prototyping and verification of the path detector.

D. Operating Theory

The operating principle rests on the differential reflectivity of black and white surfaces: a white surface reflects nearly all incident visible light, while a black surface absorbs most of it. An LED illuminates the surface beneath each sensor, and an adjacent LDR captures the reflected light. When the LED shines on a white surface, the LDR receives strong reflected light, its resistance drops, and the resulting divider voltage exceeds the comparator's reference level, causing the associated indicator LED to blink. When the LED shines on a black surface, little or no light is reflected, the LDR resistance remains high, and the indicator LED stays off. Duplicating this mechanism for a second LDR/LED pair allows the system to distinguish partial deviation (one sensor on black, one on white) from complete deviation (both sensors on white), with the latter additionally triggering an audible buzzer alert via the AND gate.

II. TECHNOLOGY AND LITERATURE SURVEY

This section reviews the key components used in the design and their relevant characteristics.

A. 555 Timer (NE555)

The NE555 is a widely used integrated circuit for timing, pulse generation, and oscillator applications, capable of producing highly stable timing pulses with a temperature stability on the order of 0.005%/°C [6]. In this project the NE555 is configured in astable mode (Fig. 3), in which the circuit has no stable output state and continuously switches between high and low, producing a square wave that is used to drive the emitter LEDs intermittently rather than continuously, reducing average current draw and mitigating the effect of ambient light drift.

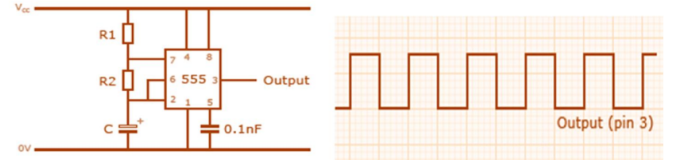


Fig. 3. Astable-mode configuration of the NE555 timer (left) and the resulting pulsed output waveform at pin 3 (right).

In astable mode, resistor R_1 is connected between V_{CC} and the discharge pin (pin 7), and resistor R_2 is connected between the discharge pin and the shared trigger (pin 2) / threshold (pin 6) node. The capacitor C charges through the series combination of R_1 and R_2 , and discharges only through R_2 , since pin 7 presents a low impedance to ground during the low portion of the output cycle. The oscillation frequency is

$$f = \frac{1}{\ln(2) \cdot C \cdot (R_1 + 2R_2)} \quad (1)$$

the high-time duration of each pulse is

$$t_{high} = \ln(2) \cdot (R_1 + R_2) \cdot C \quad (2)$$

and the low-time duration is

$$t_{low} = \ln(2) \cdot R_2 \cdot C \quad (3)$$

where R_1 and R_2 are in ohms and C is in farads. To avoid excessive dissipation, the power rating of R_1 must satisfy

$$P_{R_1} > \frac{V_{CC}^2}{R_1}. \quad (4)$$

B. Light Dependent Resistor (LDR)

An LDR, also known as a photoresistor or photoconductor (Fig. 4), is a passive component whose resistance varies inversely with incident light intensity. In darkness its resistance is very high, typically on the order of 1 MΩ, and it drops sharply under illumination. This property is exploited directly: each LDR, paired with a fixed or variable resistor as a voltage divider, converts the amount of light reflected from the surface beneath the sensor into an analog voltage that can be compared against a threshold.



Fig. 4. Light Dependent Resistor (LDR) used as the optical sensing element.

C. Comparator (LM324N)

The LM324N is a 14-pin IC (Fig. 5) containing four independent, internally compensated operational amplifiers in a single package, capable of single-supply operation from 3.0 V to 32 V [7]. Each amplifier can be configured as a comparator, oscillator, rectifier, or general-purpose amplifier. In this design, two of the four amplifiers are configured as comparators, each comparing an LDR-derived voltage at the inverting input against a fixed reference voltage at the non-inverting input, switching output state whenever the sensed voltage crosses the threshold.

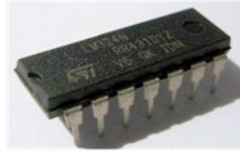
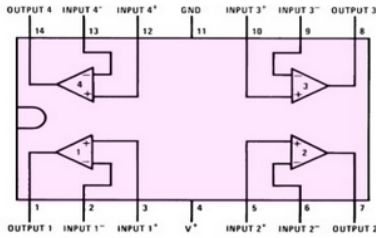


Fig. 5. LM324N pin configuration (left) and physical IC package (right).

D. AND Gate (MC74F08)

The MC74F08 is a quad 2-input AND gate IC (Fig. 6) belonging to the FAST TTL logic family, with typical propagation delays in the range of 2.5–7.5 ns [8]. Each gate produces a logic-HIGH output only when both of its inputs are logic-HIGH. In this design, a single AND gate combines the two comparator outputs so that the buzzer is activated only when both the left and right sensors simultaneously indicate a white (off-path) surface, as summarized by the truth table in Table II.

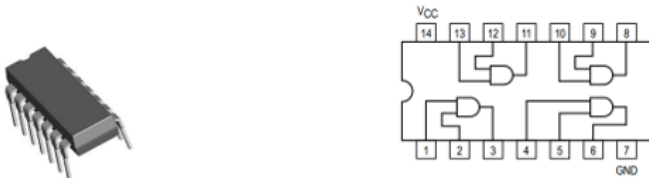


Fig. 6. 2-input AND gate logic symbol (top) and MC74F08 pin configuration (bottom right).

E. Buzzer, LED, and Potentiometer

The buzzer (Fig. 7) is an audio signaling device used to provide an audible alert confirming that the device has left the

TABLE II
SENSOR-TO-ALARM LOGIC TRUTH TABLE

Left (White?)	Right (White?)	LED State	Buzzer (AND out)	Interpretation
0	0	Both off	0	Fully on black path
1	0	Left on	0	Partial left deviation
0	1	Right on	0	Partial right deviation
1	1	Both on	1	Path fully lost

intended black path, supplementing the visual LED indicators (Fig. 8) for situations where the operator's attention is not fixed on the device. Two 100 k Ω potentiometers (Fig. 9) are used in conjunction with the LDRs to form adjustable voltage dividers, allowing the sensitivity threshold of each sensing channel to be tuned to compensate for ambient lighting conditions and surface-reflectivity variation.



Fig. 7. Piezoelectric buzzer used for the audible off-path alarm.

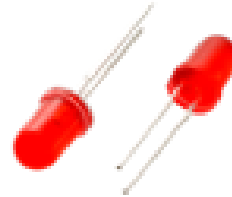


Fig. 8. Light-emitting diodes (LEDs) used as optical emitters and visual path-deviation indicators.

III. SYSTEM DESIGN AND CIRCUIT DESCRIPTION

A. Power Supply Design

A regulated logic supply of approximately +5 V (V_{CC}) is derived from a 9 V battery using a resistive voltage-drop arrangement, avoiding the cost and complexity of a dedicated linear regulator IC while remaining adequate for the low current draw of the CMOS/TTL logic and comparator stages.

B. Complete Circuit and Signal Flow

Fig. 10 shows the complete electronic circuit of the path detector. The NE555 astable stage (resistors $R3 = 1$ k Ω ,



Fig. 9. 100 kΩ potentiometer used as an adjustable sensitivity control in each LDR voltage-divider channel.

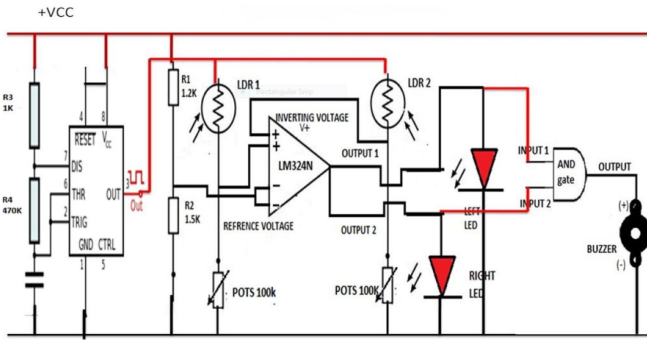


Fig. 10. Complete electronic circuit of the path detector, integrating the NE555 astable timer, dual LDR/potentiometer sensing channels, LM324N comparator stage, indicator LEDs, MC74F08 AND gate, and buzzer.

$R4 = 470 \text{ k}\Omega$) is powered between V_{CC} and ground, with its pin-3 output producing the pulsed drive voltage for the emitter LEDs. Each LDR (LDR1, LDR2) is paired with a 100 kΩ potentiometer to form a voltage divider whose midpoint serves as the inverting input to the corresponding LM324N comparator stage. A series combination of two fixed resistors ($R1 = 1.2 \text{ k}\Omega$ and $R2 = 1.5 \text{ k}\Omega$) establishes a stable non-inverting reference voltage shared by both comparators, so that the two sensing channels operate against a common, easily reproducible threshold while retaining independent sensitivity adjustment via their respective potentiometers.

C. Logic and Alarm Stage

The outputs of the two comparator stages independently drive the left-end and right-end indicator LEDs, and are also fed as the two inputs to the MC74F08 AND gate. Because the AND gate output is HIGH only when both comparator outputs are HIGH (Table II), the buzzer activates exclusively under the complete-path-loss condition, preventing nuisance alarms during normal, single-sided path correction.

IV. RESULTS AND PERFORMANCE EVALUATION

The black surface is treated as the correct path, while the white surface represents an incorrect path. Each sensing



Fig. 11. Illustration of the sensing phenomenon: low light intensity is received by the sensor over a black surface (left), while high light intensity is received over a white surface (right).

module consists of a light emitter (LED) and a light sensor (LDR); the emitter projects light onto the surface below, and any reflected light is captured by the adjacent LDR, as illustrated in Fig. 11.

When a sensor is over a black surface, most of the incident light is absorbed and very little light reaches the LDR; in this condition the corresponding warning LED does not blink. When a sensor is over a white surface, most of the incident light is reflected back to the LDR, raising the voltage delivered to the comparator. Once this voltage exceeds the reference voltage, the comparator output switches state, causing the associated LED to blink and supplying a logic-HIGH signal to the AND gate.

If only the right-hand sensor deviates onto the white surface, the right-most LED blinks; if only the left-hand sensor deviates, the left-most LED blinks. If both sensors simultaneously detect white surface – indicating that the entire device has left the black path – both LEDs blink and the AND gate output activates the buzzer, producing an audible warning tone. This graduated response (visual-only for partial deviation, visual-plus-audible for complete deviation) was verified to behave consistently and repeatably during breadboard and PCB testing on both straight and gently curved black paths on a white background.

V. DISCUSSION AND CONCLUSION

A. Discussion

As this was an introductory project for students newly exposed to practical electrical and electronics work, a substantial portion of the development time was devoted to becoming familiar with breadboard prototyping and with the circuit configuration of the timer, comparator, and logic-gate ICs, drawing on reference texts and online tutorials. Several implementation issues – including threshold drift under varying ambient light and correct biasing of the comparator reference network – were encountered during development and were resolved with guidance from the project supervisor and through online technical forums.

B. Limitations and Future Work

The present design relies on fixed-position LDR pairs and manually tuned potentiometer thresholds, which makes the system somewhat sensitive to changes in ambient lighting and to the specific reflectivity of the printed line. Future

iterations of this work could incorporate: (i) an automatic gain or threshold calibration routine, (ii) additional sensor channels to improve curve-tracking resolution, and (iii) integration with a motor-driver stage to realize a complete closed-loop line-following robot rather than a standalone detector module.

C. Conclusion

The path detector was designed and developed over a period of fourteen weeks. The project originated from an interest in developing a line-following robot, and the resulting sensing module represents a foundational step toward that broader goal; further refinement and extension of the design remain an ongoing process. Within the allocated time and resource constraints, the path detector was developed to a functional prototype stage and was demonstrated to reliably distinguish between black and white path surfaces using low-cost, readily available components, without requiring a microcontroller.

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